



IEEE VTS Motor Vehicle Challenge 2027



Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Road

The Motor Vehicle Challenge (MVC), supported by IEEE Vehicular Technology Society, is an annual activity to find an appropriate energy management strategy to improve electric vehicles' performance. In this eleventh edition of the event, the challenge focuses on the energy-efficient and safe operation of an in-wheel-motor vehicle with dynamic wireless power transfer operating on a road subject to various disturbances. The competition is open to everyone, students, academics, and industry.

DWPT CHARGING COILS

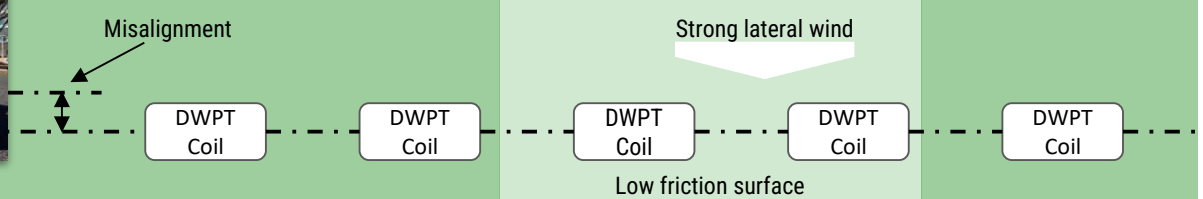
CHALLENGE

This challenge considers a four in-wheel-motor vehicle (4IWM-EV) operating autonomously on roads with a dynamic wireless power transfer (DWPT) infrastructure, based on prototypes developed at the University of Tokyo. Under realistic conditions, the vehicle is subjected to abrupt variations in road friction due to weather changes, as well as strong lateral wind disturbance. Participants are therefore invited to develop an integrated strategy that coordinates motor torque distribution, d-axis current control, and front steering angle.

The MVC2027 intends to serve as a platform for discussing and advancing the concept of MESO (Management of Energy Efficiency and Safe Operation) for multi-actuated vehicles.



4IWM-EV



SCORING

Participants of the Challenge will be provided with a model of the 4IWM-EV and DWPT platform on MATLAB/Simulink, together with sample control strategies and speed profiles.

Each proposed strategy will be evaluated using a scoring function that assesses the achievement of multiple objectives, including:

- improving overall drivetrain efficiency;
- ensuring safe traction by regulating wheel slip ratios;
- enhancing infrastructure utilization by maximizing the energy received from DWPT;
- improving driving comfort by minimizing deviations in lateral acceleration.

PRIZES

Teams that develop the best solutions will receive **an award** consisting of a certificate, an invitation to write and present a paper for the IEEE VPPC 2027, and **a grant** that supports the participation in the IEEE VPPC 2027:

- **First prize:** Up to a limit of 3500 US\$.
- **Second prize:** Up to a limit of 1500 US\$.

Open to everyone; only IEEE VTS members are eligible for the grant.

MATERIAL AND REGISTRATION

More information, registration and material is available on:

- [\[https://github.com/VTSociety/MVC_2027\]](https://github.com/VTSociety/MVC_2027)
- [Registration form](#)

IMPORTANT DATES

- Registration deadline: 31 January 2027
- Submission: 1 March 2027
- Results: 31 March 2027

IEEE VTS MVC 2027 ORGANIZING COMMITTEE

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